

```
Open  p1.c  Save  x
/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01

1#include<stdio.h>
2#include<unistd.h>
3#include<sys/wait.h>
4#include<stdlib.h>
5
6int main() {
7    pid_t pid;
8    char command[50];
9
10   printf("Enter command: ");
11   scanf("%49s", command);
12
13   printf("Parent process: \n");
14   printf("PID: %d\n", getpid());
15   printf("PPID: %d\n", getppid());
16
17   pid = fork();
18
19   if (pid < 0) {
20       printf("Fork() invocation failed\n");
21       exit(1);
22   } else if (pid == 0) {
23       printf("Child Process: %d\n", getpid());
24       printf("PID: %d\n", getpid());
25       printf("PPID: %d\n", getppid());
26
27       execlp(command, command, NULL);
28
29       printf("exec() failed\n");
30       exit(1);
31   } else {
32       printf("Parent Process\n");
33       wait(NULL);
34       printf("Parent Resumed\n");
35       printf("Child process finished\n");
36       exit(0);
37   }
38   return 0;
39 }
```

```
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ gcc p1.c
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ ./a.out
Enter command: ls
Parent process:
PID: 4452
PPID: 321
Parent Process
Child Process: 4453
PID: 4453
PPID: 4452
a.out p1.c
Parent Resumed
Child process finished
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ ./a.out
Enter command: ls -l
Parent process:
PID: 4463
PPID: 321
Parent Process
Child Process: 4464
PID: 4464
PPID: 4463
a.out p1.c
Parent Resumed
Child process finished
```

```
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ uname
Linux
```

```
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-1355U
CPU family:             6
Model:                 186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:               5222.40
Flags:                  fpu vme de pse tsc mtr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr
                        sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_re
                        liable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse
                        4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm
                        abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase
                        tsc_adjust bni1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsav
                        eopt xsavec xgetby1 xsavec avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri
                        movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    288 KiB (6 instances)
L1i:                    192 KiB (6 instances)
L2:                     7.5 MiB (6 instances)
L3:                     12 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-11
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
Itlb multihit:          Not affected
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:         Not affected
Old microcode:           Not affected
Reg file data sampling:  Mitigation; Clear Register File
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI BH
                        I_OIS_S
Srbds:                  Not affected
Tsx:                    Not affected
Tsx async abort:         Not affected
Vmscape:                 Not affected
```

```
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
```

```
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ ps
PID TTY          TIME CMD
321 pts/0        00:00:00 bash
4798 pts/0        00:00:00 ps
```

```
dell@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ top
top - 06:49:11 up 10 min, 1 user, load average: 0.18, 0.15, 0.06
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.7 sy,  0.0 ni, 99.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7784.6 total, 7072.2 free,  589.3 used,  276.0 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used,  7195.3 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 24340 15320 11512 S   0.0   0.2   0:00.74 systemd
    2 root        20   0 3180 2208 2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root        20   0 3180 2048 1940 S   0.0   0.0   0:00.00 init
   46 root       19  -1 50404 16920 15608 S   0.0   0.2   0:00.23 systemd-journal
   78 systemd+   20   0 22416 14492 11992 S   0.0   0.2   0:00.08 systemd-resolve
   85 root        20   0 35364 12296 9216 S   0.0   0.2   0:00.40 systemd-udev
  134 root        20   0 2880 1948 1520 S   0.0   0.0   0:00.05 chronyd-starter
  139 root        20   0 4472 3116 2868 S   0.0   0.0   0:00.01 cron
  145 message+   20   0 8824 5384 4684 S   0.0   0.1   0:00.07 dbus-daemon
  152 root        20   0 44096 29772 14624 S   0.0   0.4   0:00.27 networkd-dispat
  156 root        20   0 18496 9504 8204 S   0.0   0.1   0:00.12 systemd-logind
  159 root        20   0 1866288 15320 12552 S   0.0   0.2   0:00.12 wsl-pro-service
  178 syslog      20   0 220548 5960 4656 S   0.0   0.1   0:00.08 rsyslogd
  216 _chrony      20   0 23688 11112 7076 S   0.0   0.1   0:00.09 chronyd
  219 root        20   0 5492 2732 2548 S   0.0   0.0   0:00.00 agetty
  225 _chrony      20   0 12096 2340 1696 S   0.0   0.0   0:00.01 chronyd
  269 root        20   0 123460 32596 17988 S   0.0   0.4   0:00.24 unattended-upgr
  319 root        20   0 3196 1116 900 S   0.0   0.0   0:00.00 SessionLeader
  320 root        20   0 3196 1252 1100 S   0.0   0.0   0:00.00 Relay(321)
  321 dell        20   0 6396 5732 3928 S   0.0   0.1   0:00.15 bash
  322 root        20   0 8644 5644 4804 S   0.0   0.1   0:00.01 login
  360 dell        20   0 22512 13296 10840 S   0.0   0.2   0:00.18 systemd
  360 dell        20   0 22912 4100 2100 S   0.0   0.1   0:00.00 (sd-pam)
  394 dell        20   0 6136 5400 3864 S   0.0   0.1   0:00.08 bash
```